

MACHINE LEARNING FOR BI

PART 1:

QA & DATA PROFILING



★★★★★ *With Expert ML Instructor Josh MacCarty*



ABOUT THIS SERIES

This is **Part 1** of a **4-Part series** designed to help you build a deep, foundational understanding of machine learning, including data QA & profiling, classification, forecasting and unsupervised learning



PART 1

QA & Data Profiling



PART 2

Classification



PART 3

Regression & Forecasting



PART 4

Unsupervised Learning

COURSE OUTLINE

1	ML Intro & Landscape	<i>Machine Learning introduction, definition, process & landscape</i>
2	Preliminary Data QA	<i>Tools to explore data quality (variable types, empty values, range & count calculations, table structure, left/right censored data, etc.)</i>
3	Univariate Profiling	<i>Tools to understand individual variables (distribution, histograms & kernel densities, data profiling metrics, etc.)</i>
4	Multivariate Profiling	<i>Tools to understand multiple variables (kernel densities, violin & box plots, correlation, variance, etc.)</i>

SETTING EXPECTATIONS



This is **NOT** a coding/programming course; it's about introducing and demystifying essential machine learning topics

- *Our goal is to break down complex techniques using simple and intuitive explanations and demos*



We'll focus on tools commonly applied to **business intelligence** use cases

- *We'll focus on techniques like data profiling, linear/logistic regression, forecasting, and unsupervised learning, but will not cover some more advanced or specialized techniques (deep learning, NLP, etc.)*



We'll use **Microsoft Excel** as a tool to help explain key concepts

- *Excel's intuitive, visual interface allows us to expose the nuts and bolts of each technique to understand HOW and WHY these algorithms work (rather than simply running lines of code)*



You do **NOT** need a math or stats background to take this course

- *We'll cover the basics as needed, but won't dive deep into statistics or econometric theory*

SETTING EXPECTATIONS



Who this is for:

- Analysts or BI professionals looking to transition into a ML/data science role
- Students looking to develop a deep conceptual understanding of core machine learning topics
- Anyone who wants to understand WHEN, WHY, and HOW to deploy machine learning tools & techniques



Who this is NOT for:

- Senior data professionals looking to master advanced topics in ML/AI
- Students looking for a hands-on coding course or bootcamp (i.e. python/R)
- Anyone who would rather copy and paste code than become fluent in the underlying algorithms

ML INTRO & LANDSCAPE

INTRO TO MACHINE LEARNING (ML)



MACHINE LEARNING [*muh-sheen-lur-ning*]

noun

- ~~1. The capacity of a computer to process and evaluate data beyond programmed algorithms, through contextualized inference*~~



Using statistical models to find patterns and make predictions

COMMON ML QUESTIONS



Which customers are most likely to churn next month?



What will sales look like for the next 12 months?



What patterns do we see in terms of product cross-selling?



How can we use online customer reviews to monitor changes in sentiment?

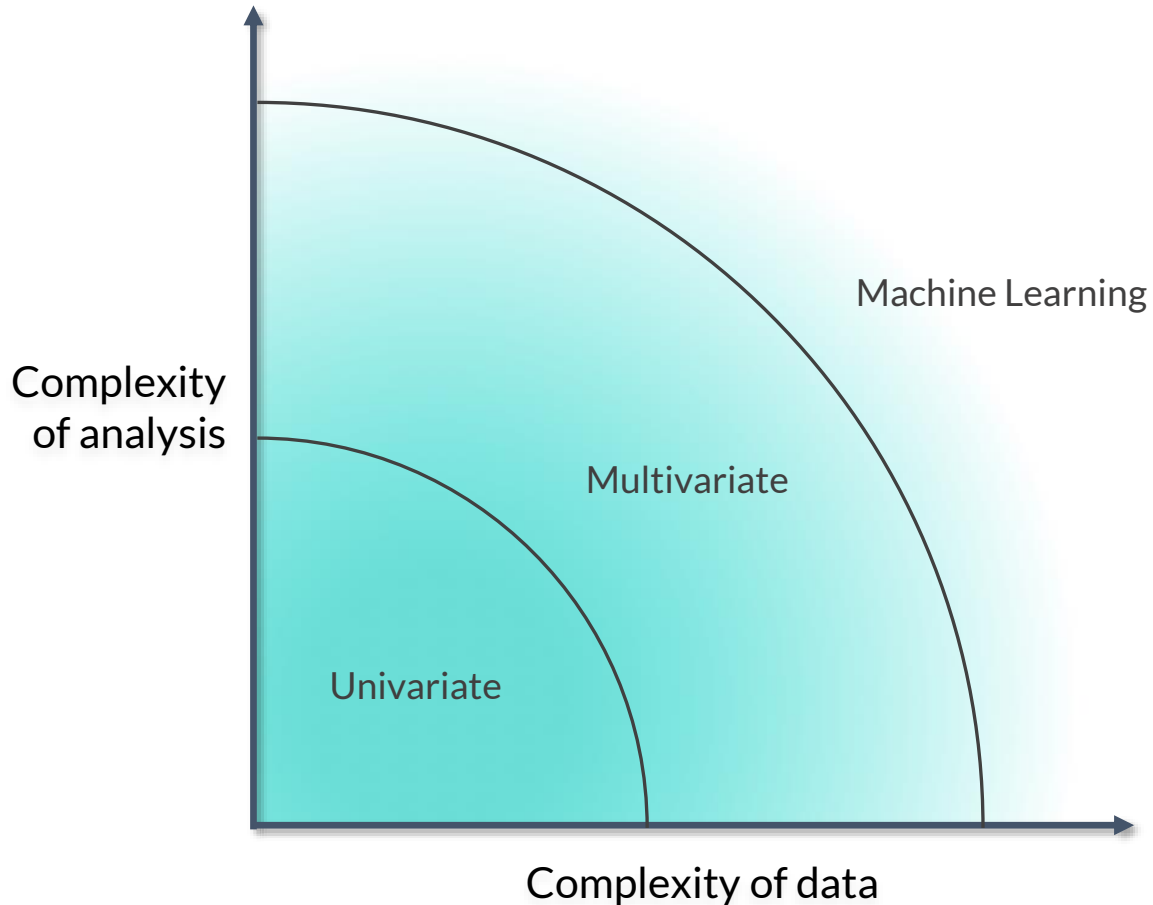


When we adjusted tactics last month, did we drive any incremental revenue?



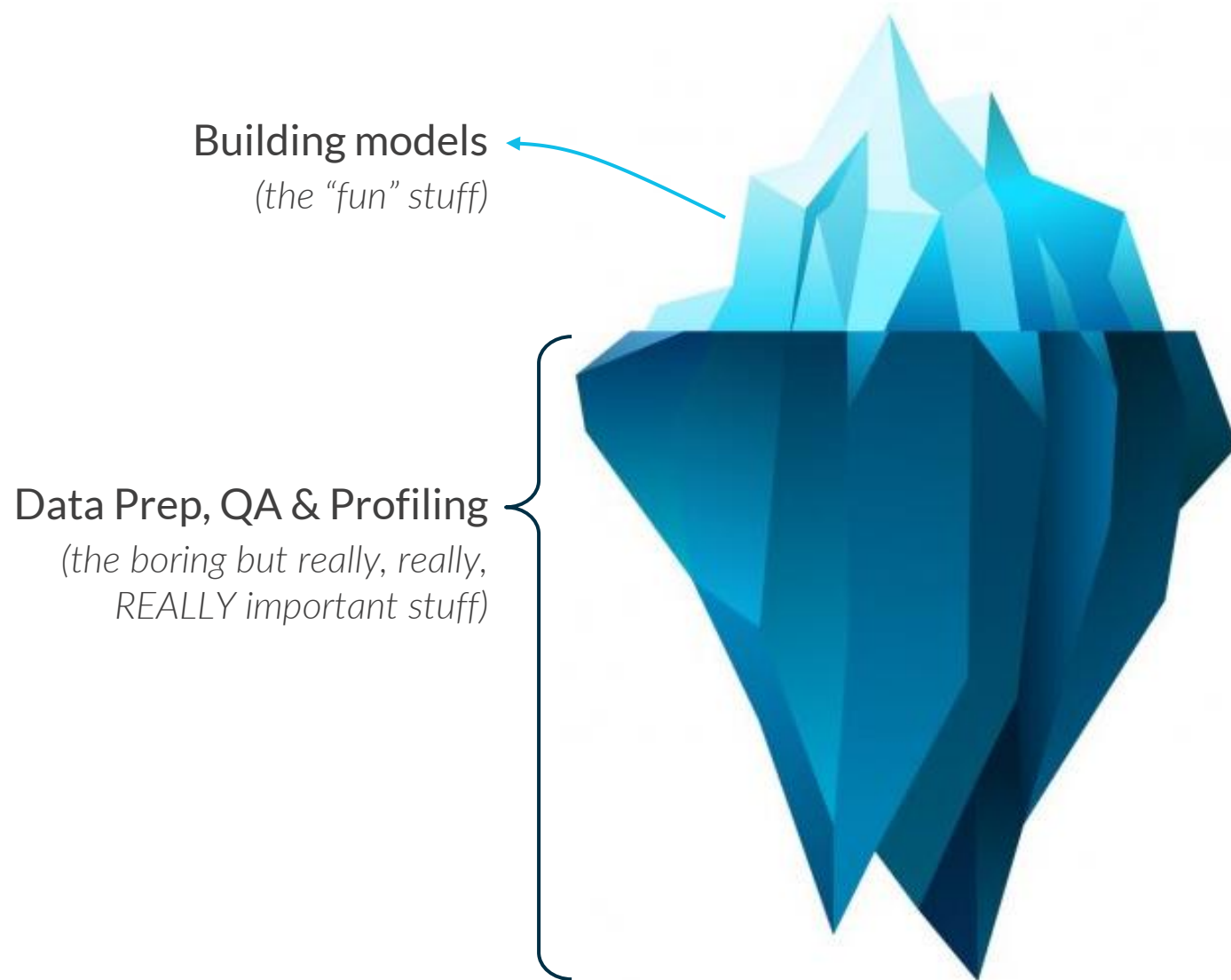
Which product is customer X most likely to purchase next?

WHEN IS ML THE RIGHT FIT?



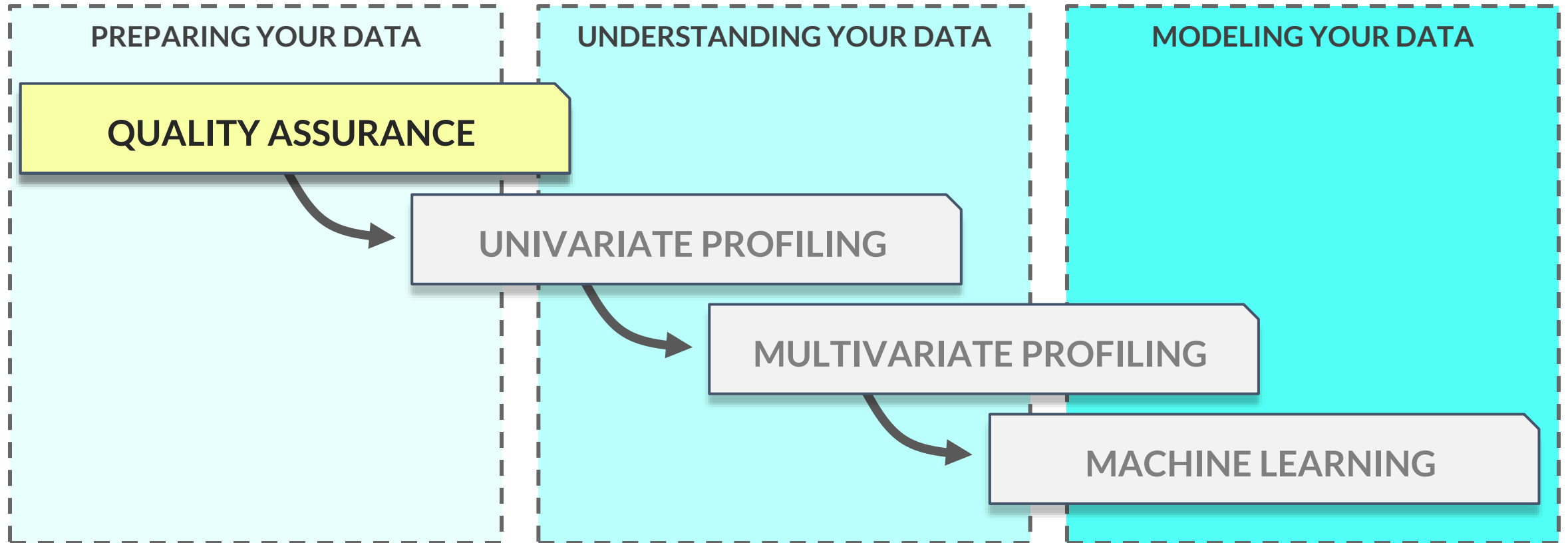
- Machine Learning is a natural extension of data profiling and basic visual analysis
- ML is required when the underlying data or analysis is **too complex for basic data profiling** (*i.e. visualizing relationships between 3+ variables*)
- Machine Learning is ideal for finding optimal solutions that would be **impossible or impractical to derive through human trial-and-error**

THE MACHINE LEARNING PROCESS



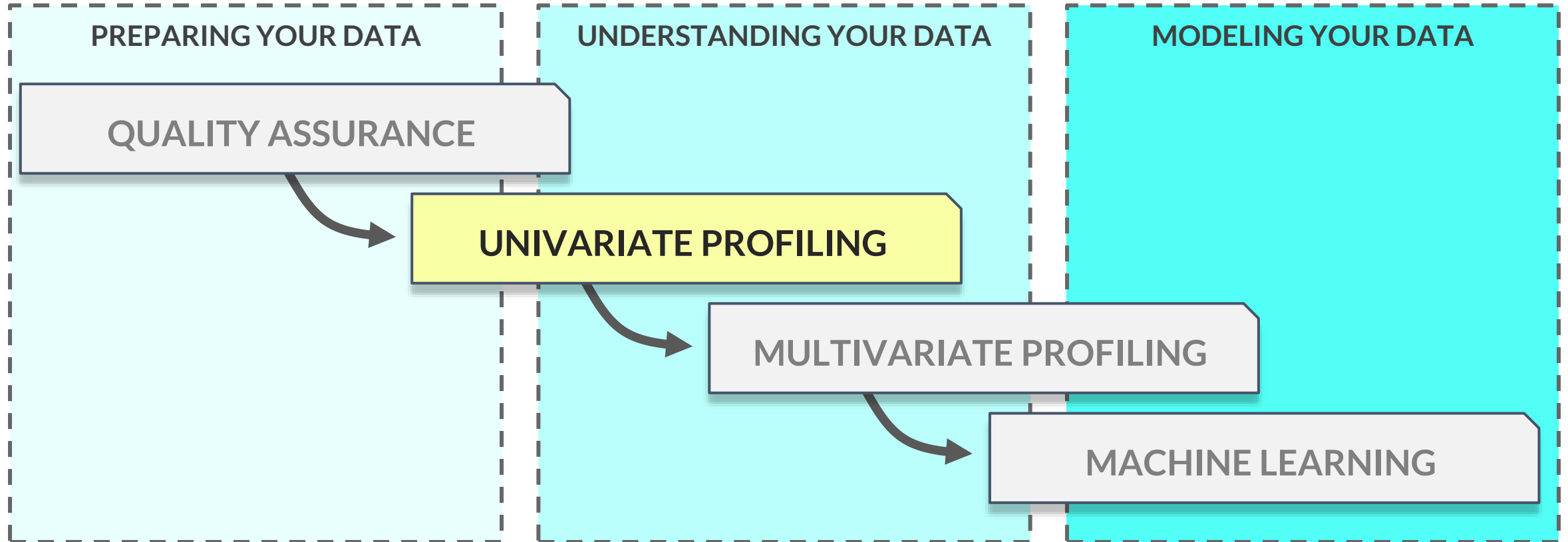
- ML models are only as good as the data they are built on (“garbage in, garbage out”)
- Data prep & QA happens behind the scenes, but typically accounts for the majority (**80%+**) of the machine learning workflow
- While it’s tempting to start with the “fun” stuff, data prep and QA skills are ***absolutely critical!***

MACHINE LEARNING PROCESS



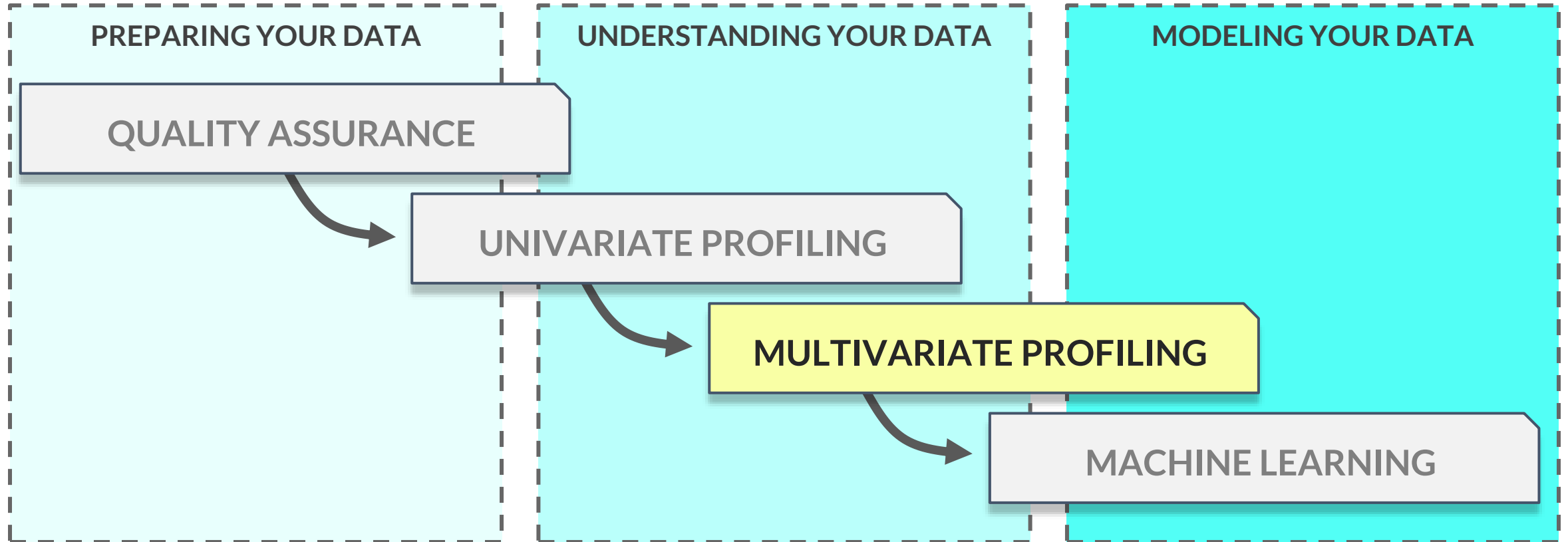
Quality Assurance (QA) is about **preparing & cleaning data prior to analysis**. We'll cover common QA topics including variable types, empty/missing values, range & count calculations, censored data, etc.

MACHINE LEARNING PROCESS



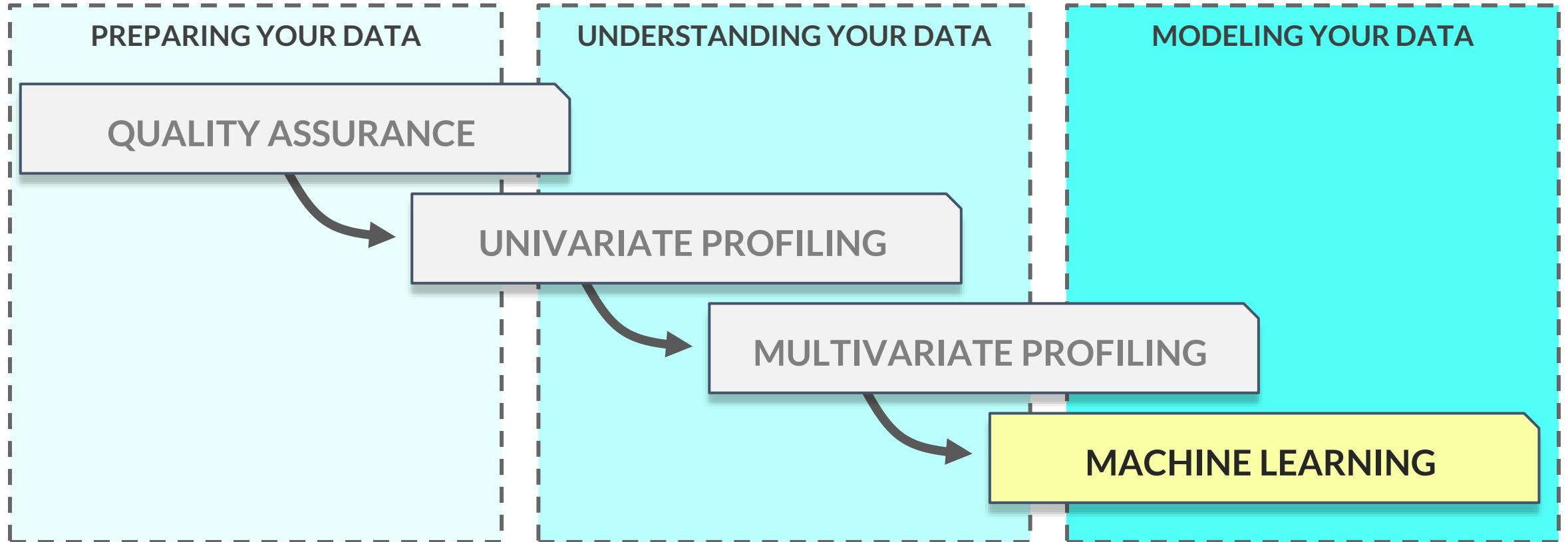
Univariate profiling is about **exploring individual variables** to build an understanding of your data. We'll cover common topics like normal distributions, frequency tables, histograms, etc.

MACHINE LEARNING PROCESS



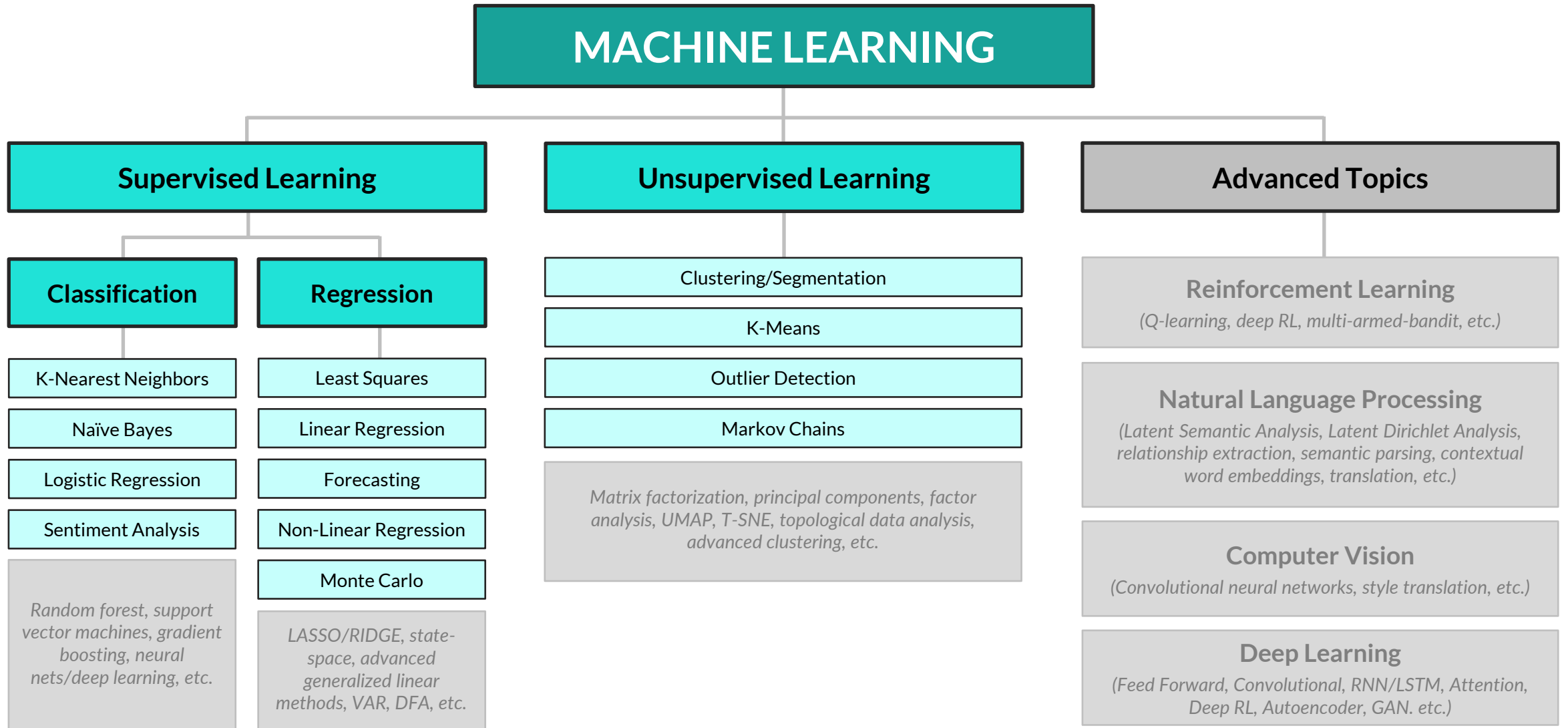
Multivariate profiling is about **understanding relationships between *multiple* variables**. We'll cover common tools for exploring categorical & numerical data, including kernel densities, violin & box plots, scatterplots, etc.

MACHINE LEARNING PROCESS



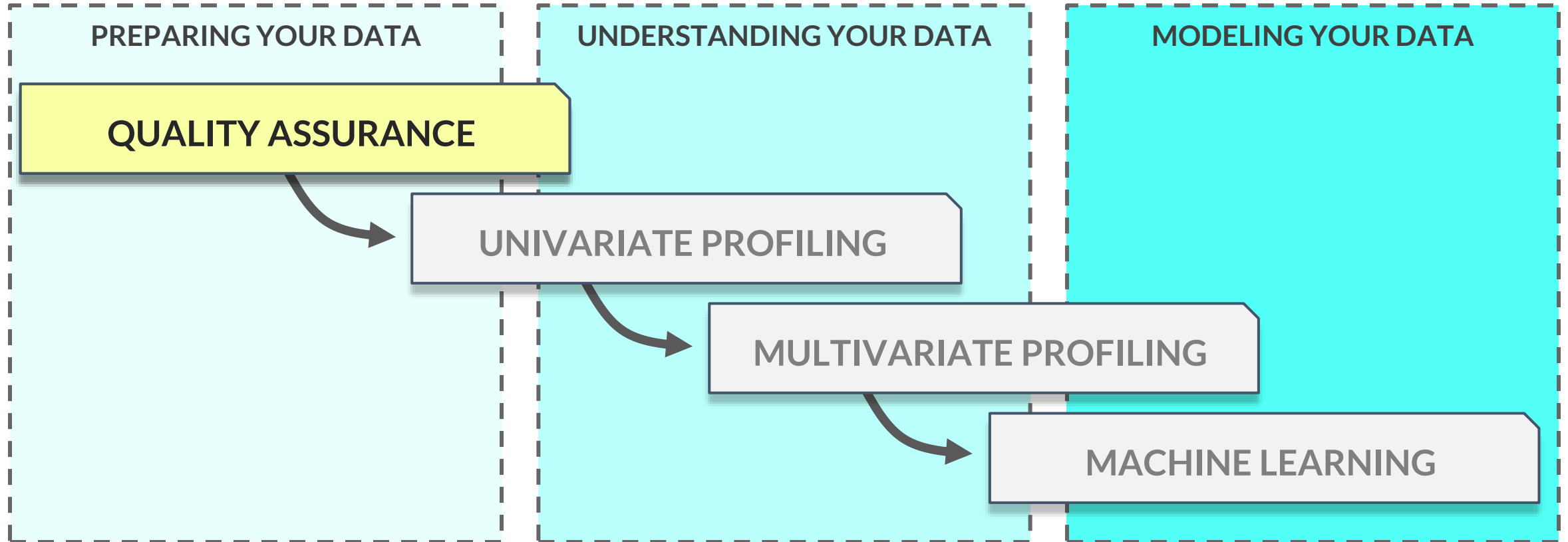
Machine learning is a natural extension of multivariate profiling, and uses **statistical models and methods** to answer questions which are too complex to solve using simple visual analysis or trial-and-error

MACHINE LEARNING LANDSCAPE



PRELIMINARY DATA QA

MACHINE LEARNING PROCESS



Quality Assurance (QA) is about **preparing & cleaning data prior to analysis**. We'll cover common QA topics including variable types, empty/missing values, range & count calculations, censored data, etc.

PRELIMINARY DATA QA



Data QA (otherwise known as **Quality Assurance** or **Quality Control**) is the first step in the analytics and machine learning process; QA allows you to identify and correct underlying data issues (*blanks, errors, incorrect formats, etc.*) prior to analysis

TOPICS WE'LL COVER:

Variable Types

Empty Values

Range Calculations

Count Calculations

Table Structures

Left/Right Censors

COMMON USE CASES:

- Minimizing the risk of drawing false conclusions or misunderstanding your data
- Confirming that dates are properly formatted for analysis (*as date values, not text strings*)
- Replacing blanks or errors with appropriate values to prevent summarization errors
- Calculating ranges (*max & min*) to spot-check for outliers or unexpected values

PRELIMINARY DATA QA

💡 **POP QUIZ:** *When should you QA your data?*

EVERY. SINGLE. TIME.

(no exceptions!)

WHY IS QA IMPORTANT?

As an analyst, **Preliminary Data QA** will help you answer questions like:



Are there any missing or empty values in the data shared by the HR team?



Was the data our client captured from the online survey encoded properly?



Is there any risk that the data capture process was biased in some way?



Are there any outliers that might skew the results of our analysis?

VARIABLE TYPES

Variable Types

Empty Values

Range Calculations

Count Calculations

Left/Right Censored

Table Structure

Variable types give us information about our variables

- January 1, 2000 as a **number** simply displays a date
- January 1, 2000 as a **date** implies a set of information we can use (*first day of the year, winter, Saturday, weekend, etc.*) all of which can be used to build strong machine learning models

Common **variable types** include:

Numeric

- Customer count

Discrete

- Count

Date

- 1/1/2020

Ordinal

- Small, Medium, Large

Categorical

- Gender

Complex

- $4 + 2i$

Interval

- Temperature

Nominal

- Nationality

Logical

- True/False

Ratio

- Weight

Binary

- Yes/No

Monetary

- \$4.50

VARIABLE TYPES

Variable Types

Empty Values

Range Calculations

Count Calculations

Left/Right Censored

Table Structure

Understanding **variable types** is fundamental to the machine learning process, and can help avoid common QA issues, including:

- Numeric variables formatted as **characters** or **strings**, preventing proper aggregation or analysis
- **String/character** variables formatted as **numeric**, preventing proper text-based operations
- Values that Less obvious cases like variables that are re-coded from raw values to buckets (surveys)

ZIP CODES	
Numeric	String
90210	90210
32992	32992
1730	01730
2215	02215
92120	92120

Here we're formatting **zip codes** (which will never be analyzed as values), as a **numeric** rather than **string**



When we talk about **variables**, we might also refer to them as **metrics**, **KPIs**, **dimensions**, or **columns**. Similarly, we may use **rows**, **observations**, and **records** interchangeably

EMPTY VALUES

Variable Types

Empty Values

Range Calculations

Count Calculations

Left/Right Censored

Table Structure

Investigating **empty values**, and how they are recorded in your data, is a prerequisite for *every single* analysis

Empty values can be recorded in many ways (NA, N/A, #N/A, NaN, Null, "-", "Invalid", blank, etc.), but the most common mistake is **turning empty numerical values into zeros (0)**

Sales Leaders		
Name	Age	Sales
John	33	278
Sally	30	173
Mark	38	210
Gina	0	122
Phil	42	246

Average Age = 28.6 ❌

Sales Leaders		
Name	Age	Sales
John	33	278
Sally	30	173
Mark	38	210
Gina		122
Phil	42	246

Average Age = 35.8 ✅

EMPTY VALUES

Variable Types

Empty Values

Range Calculations

Count Calculations

Left/Right Censored

Table Structure

Empty values can be handled in 3 ways: **keep**, **remove**, or **impute**

- **Keep** empty or zero values if you are certain that they are accurate and meaningful (*i.e. no sales of a product on a specific date*)
- **Remove** empty values if you have a large volume of data and can confirm that there is no pattern or bias to the missing values (*i.e. all sales from a specific product category are missing*)
- **Impute** (substitute) empty values if you can accurately populate the data or if you are working with limited data and can use statistical methods (*mean, conditional mean, linear interpolation, etc.*) without introducing bias

PRODUCT SALES				
Date	Product Name	Product ID	Retail Price	Units Sold
5-Jan-20	Club Chocolate Bar	10660	\$1.35	4
5-Jan-20	Big Time Waffles	62736	\$3.69	
6-Jan-20	Excellent Apple Drink	30389	\$2.98	7
7-Jan-20	High Top Oranges	98772		11
7-Jan-20	Jeffers Oatmeal	60396	\$1.54	3

For a missing Units Sold value, you would likely **remove** the row unless you are certain that an empty value represents 0 sales

For a missing Retail Price, you would likely be able to **impute** the value since you know the product name/ID

RANGE CALCULATIONS

Variable Types

Empty Values

Range Calculations

Count Calculations

Left/Right Censored

Table Structure

One of the simplest QA tools for numerical variables is to calculate the **range of values** in a column (*minimum and maximum values*)

Range calculation is a helpful tool to understand your variables, confirm that ranges are realistic, and identify potential outliers

min(Age) = **18**
max(Age) = **65**

Is there a clear **lower or upper limit**
(i.e. 18+ or capped at 65)?

min(Income) = **0**
max(Income) = **100**

Is your variable transformed to a **standard max/min scale** (i.e. 1-10, 0-100)?

min(height) = **-10**
max(height) = **10**

Is your variable **normalized around a central value** (i.e. 0)?

COUNT CALCULATIONS

Variable Types

Empty Values

Range Calculations

Count Calculations

Left/Right Censored

Table Structure

Count calculations help you understand the number of records or observations that fall within specific categories, and can be used to:

- Identify categories you were not expecting to see
- Begin understanding how your data is distributed
- Gather knowledge for building accurate ML models (*more on this later*)

Distinct counts can be particularly useful for QA, and help to:

- Understand the granularity or “grain” of your data
- Identify how many unique values a field contains
- Ensure consistency by identifying misspellings or categorization errors which might otherwise be difficult to catch (*i.e. leading or trailing spaces*)



PRO TIP: For numerical variables with many unique values (*i.e. long decimals*), use a **histogram** to plot frequency based on custom ranges or “**bins**” (*more on that soon!*)

LEFT/RIGHT CENSORED

Variable Types

Empty Values

Range Calculations

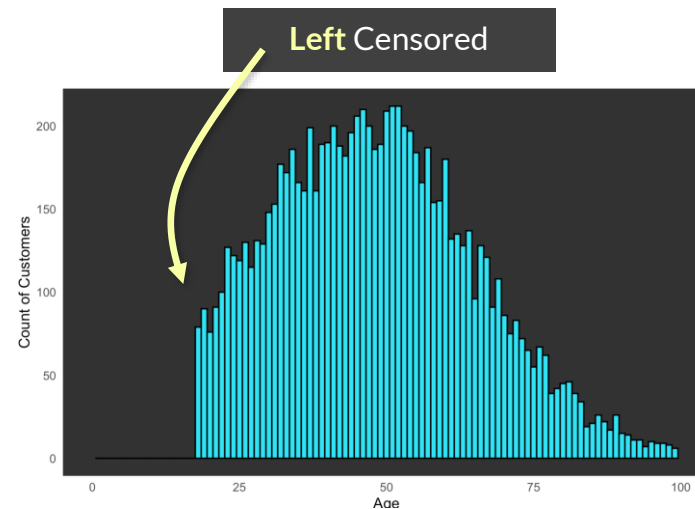
Count Calculations

Left/Right Censored

Table Structure

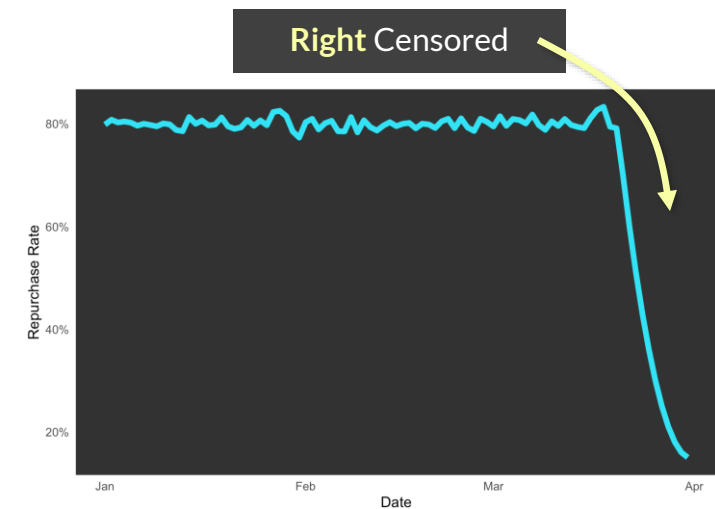
When data is **left or right censored**, it means that due to some circumstance the min or max value observed is **not the natural minimum or maximum** of that metric

- This can be difficult to spot unless you are aware of how the data is being recorded (*which means it's a particularly dangerous issue to watch out for!*)



Mall Shopper Survey Results

Only tracks shoppers over the age of 18 due to legal reasons, so anyone under 18 is excluded (even though there are plenty of mall shoppers under 18)



Ecommerce Repeat Purchase Rate

Sharp drop as you approach the current date has nothing to do with customer behavior, but the fact that recent customers haven't have the opportunity or need to repurchase yet

TABLE STRUCTURE

Variable Types

Empty Values

Range Calculations

Count Calculations

Left/Right Censored

Table Structure

Table structures generally come in two flavors: **long** or **wide**

Long Table

Product Details				
ID	Size	Section	Price	Quantity
1	L	Camping	5	2000
2	L	Camping	10	1600
3	L	Camping	30	1100
4	L	Camping	100	500
5	L	Camping	200	20
6	L	Camping	15	1700
7	L	Camping	20	1600
8	L	Camping	25	1500
9	L	Biking	600	100
10	L	Biking	50	200
11	S	Camping	4	1000
12	S	Camping	8	1000
13	S	Camping	26	600
14	S	Camping	80	900
15	S	Camping	190	10
16	S	Camping	12	500
17	S	Biking	600	150
18	S	Biking	50	1000
19	S	Biking	20	300
20	S	Biking	30	700

Wide Table

Product Details				
Section	Large Price	Small Price	Large Quantity	Small Quantity
Camping	400	300	10	20
Biking	200	150	100	110

PIVOT

UNPIVOT



Pivoting is the process of adjusting a table from long to wide by transforming **rows** into **columns**, and **Unpivoting** is the opposite (*wide to long*)

TABLE STRUCTURE

Variable Types

Empty Values

Range Calculations

Count Calculations

Left/Right Censored

Table Structure

Long tables typically contain a **single, distinct column** for each field (*Date, Product, Category, Quantity, Profit, etc.*)

- Easy to see all available fields and variable types
- Great for exploratory data analysis and aggregation (*i.e. PivotTables*)

Wide tables typically split the same metric into **multiple columns** or categories (*i.e. 2018 Sales, 2019 Sales, 2020 Sales, etc.*)

- Typically not ideal for human readability, since wide tables may contain thousands of columns (*vs. only a handful if pivoted to a long format*)
- Often (*but not always*) the best format for machine learning model input
- Great format for visualizing categorical data (*i.e. sales by product category*)



There's no **right** or **wrong** table structure; each type has strengths & weaknesses!

CASE STUDY: PRELIMINARY QA



THE SITUATION

You've just been hired as a Data Analyst for **Maven Market**, a local grocery store looking for help with basic data management and analysis.



THE ASSIGNMENT

The store manager would like you to conduct some analyses on product inventory and sales, but the data is a mess.

You'll need to **explore the data**, **conduct a preliminary QA**, and **help clean it up** to prepare the data for further analysis.



THE OBJECTIVES

1. Look at common data profiling metrics to identify potential issues
2. Take note of any issues you find in the product inventory sample
3. Correct the issues to prepare the data for further analysis

BEST PRACTICES: PRELIMINARY QA



Review all fields to ensure that **variable types** are configured for proper analysis (*i.e. no dates formatted as strings, text formatted as values, etc.*)



Remember that **NA** and **0** do *not* mean the same thing! Think carefully about how to handle missing data and the impact it may have on your analysis



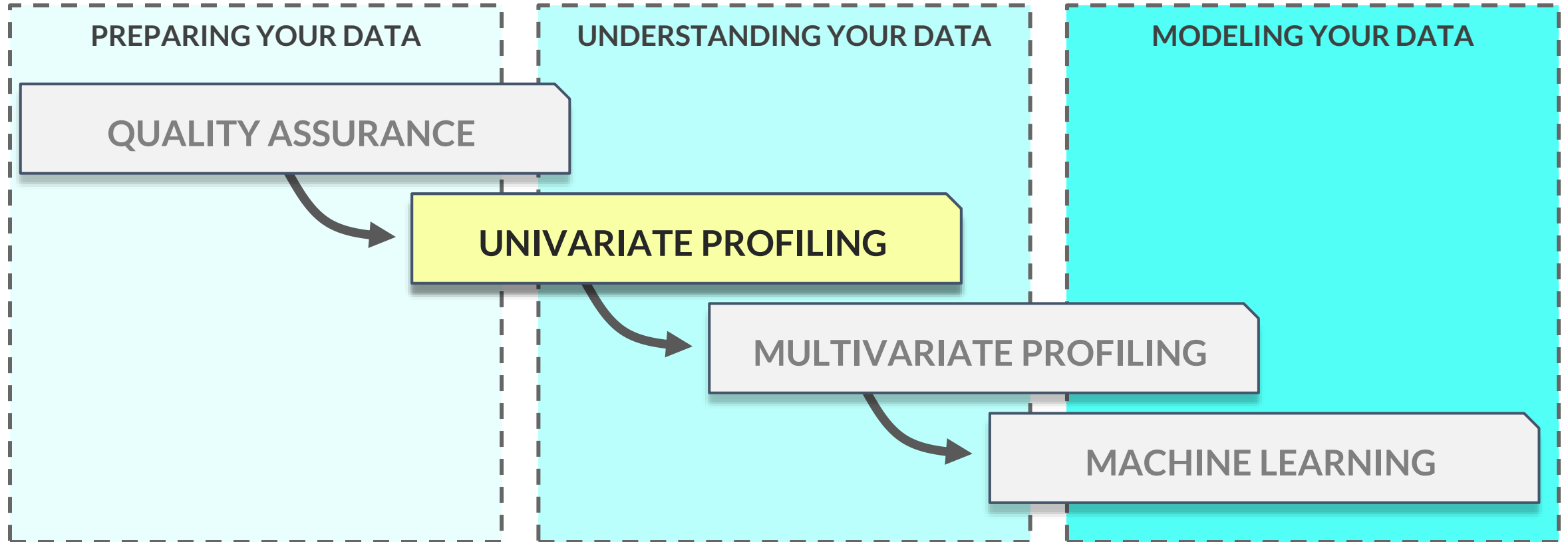
Run basic diagnostics like **Range**, **Count**, and **Left/Right Censored** checks against all columns in your data set...*every time*



Understand your **table structure** *before* conducting any analysis to reduce the risk of double counting, inaccurate calculations, omitted data, etc.

UNIVARIATE PROFILING

MACHINE LEARNING PROCESS



Univariate profiling is about **exploring individual variables** to build an understanding of your data. We'll cover common topics like normal distributions, frequency tables, histograms, etc.

UNIVARIATE PROFILING



Univariate profiling is the next step after preliminary QA; think of univariate profiling as conducting a descriptive analysis of each variable *by itself*

TOPICS WE'LL COVER:

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms & Kernel
Densities

Normal Distribution

Data Profiling

COMMON USE CASES:

- Developing a deeper understanding of the fields you're working with (*outliers, distribution, etc.*)
- Preparing to build ML models (*necessary for selecting the correct model type*)
- Exploring individual variables before conducting a deeper multivariate analysis

CATEGORICAL VARIABLES

Categorical Variables

Categorical
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Kernel Densities

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Data Profiling

VARIABLE TYPES

CATEGORICAL

NUMERICAL

Categorical variables contain *categories* as values, instead of numbers (i.e. *Product Type, Customer Name, Country, Month, etc.*)

- Categorical fields are exceptionally important for data analysis, and are typically used as dimensions by which we **filter** or “**cut**” numerical values (i.e. *sales by store*)
- Many types of predictive models are used to predict categorical dependent variables, or “classes” (*more on that later!*)



Terms like **discrete**, **categorical**, **multinomial**, and **classes** may all be used interchangeably. **Binary** is a special type of categorical variable which takes only 1 of 2 cases: *true* for *false* (or 1 or 0) and is also known as a **logical** variable or a **binary flag**

DISCRETIZATION

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

Normal Distribution

Data Profiling

Discretization is the process of creating a new *categorical* variable from an existing *numerical* variable, based on the values of the numerical variable

Product Details					
ID	Size	Section	Price	Quantity	Discretized Price
1	L	Camping	5	2000	Low
2	L	Camping	10	1600	Low
3	L	Camping	30	1100	Low
4	L	Camping	100	500	Med
5	L	Camping	200	20	Med
6	L	Camping	15	1700	Low
7	L	Camping	20	1600	Low
8	L	Camping	25	1500	Low
9	L	Biking	600	100	High
10	S	Biking	50	200	Low
11	S	Camping	4	1000	Low
12	S	Camping	8	1000	Low
13	S	Camping	26	600	Low
14	S	Camping	80	900	Low
15	S	Camping	190	10	Med
16	S	Camping	12	500	Low
17	S	Biking	600	150	High
18	S	Biking	50	1000	Low
19	S	Biking	20	300	Low
20	S	Biking	30	700	Low

Discretization Rules:

If Price <100 then Price Level = Low

If Price >=100 & Price <500 then Price Level = Med

If Price >=500 then Price Level = High

NOMINAL VS. ORDINAL VARIABLES

Categorical Variables

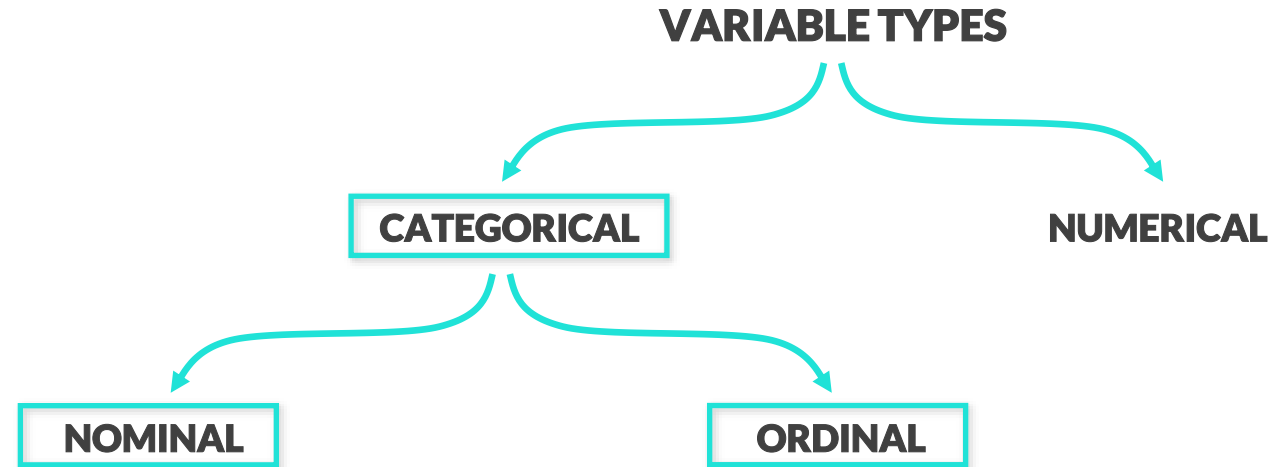
Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

Normal Distribution

Data Profiling



There are two types of categorical variables: **nominal** and **ordinal**

- **Nominal variables** contain categories with no inherent logical rank, which can be re-ordered with no consequence (*i.e. Product Type = Camping, Biking or Hiking*)
- **Ordinal variables** contain categories with a logical order (*i.e. Size = Small, Medium, Large*), but the interval between those categories has no logical interpretation

CATEGORICAL DISTRIBUTIONS

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

Normal Distribution

Data Profiling

Categorical distributions are visual and/or numeric representations of the unique values a variable contains, and how often each occurs

Common categorical distributions include:

- **Frequency tables:** Show the count (*or frequency*) of each distinct value
- **Proportions tables:** Show the count of each value as a % of the total
- **Heat maps:** Formatted to visualize patterns (*typically used for multiple variables*)

Understanding categorical distributions will help us gather knowledge for building accurate machine learning models (*more on this later!*)

CATEGORICAL DISTRIBUTIONS

Categorical Variables

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Normal Distribution

Data Profiling

Product Details				
ID	Size	Section	Price	Quantity
1	L	Camping	5	2000
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3	L	Camping	30	1100
4	L	Camping	100	500
5	L	Camping	200	20
6	L	Camping	15	1700
7	L	Camping	20	1600
8	L	Camping	25	1500
9	L	Biking	600	100
10	L	Biking	50	200
11	S	Camping	4	1000
12	S	Camping	8	1000
13	S	Camping	26	600
14	S	Camping	80	900
15	S	Camping	190	10
16	S	Camping	12	500
17	S	Biking	600	150
18	S	Biking	50	1000
19	S	Biking	20	300
20	S	Biking	30	700

Section Distribution:

Camping	Biking	}	Frequency table
14	6		
Camping	Biking	}	Proportions table
70%	30%		

Size & Section Distribution:

	Camping	Biking	}	Heat Map
S	6	4		
L	8	2		

NUMERICAL VARIABLES

Categorical Variables

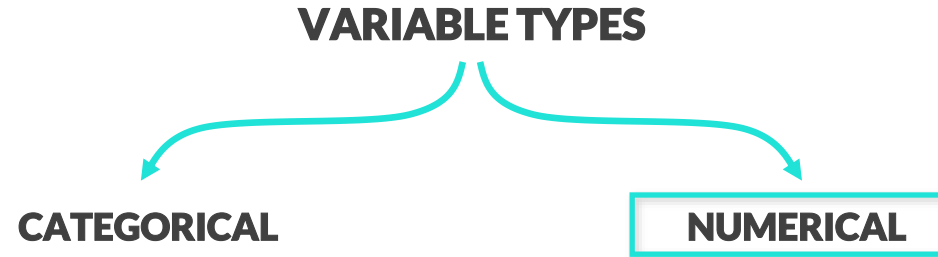
Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

Normal Distribution

Data Profiling



Numerical variables contain *numbers* as values, instead of categories (i.e. *Gross Revenue, Pageviews, Quantity, Retail Price, etc.*)

- Numerical fields are typically **aggregated** (as a *sum, count, average, max, min, etc.*) and broken down by different dimensions or categories
- Numerical data is also known as **quantitative** data, while categorical data is often referred to as “qualitative”



You may hear numeric variables described further as **interval** and **ratio**, but the distinction is trivial and rarely makes a difference in common use cases

HISTOGRAMS

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

Normal Distribution

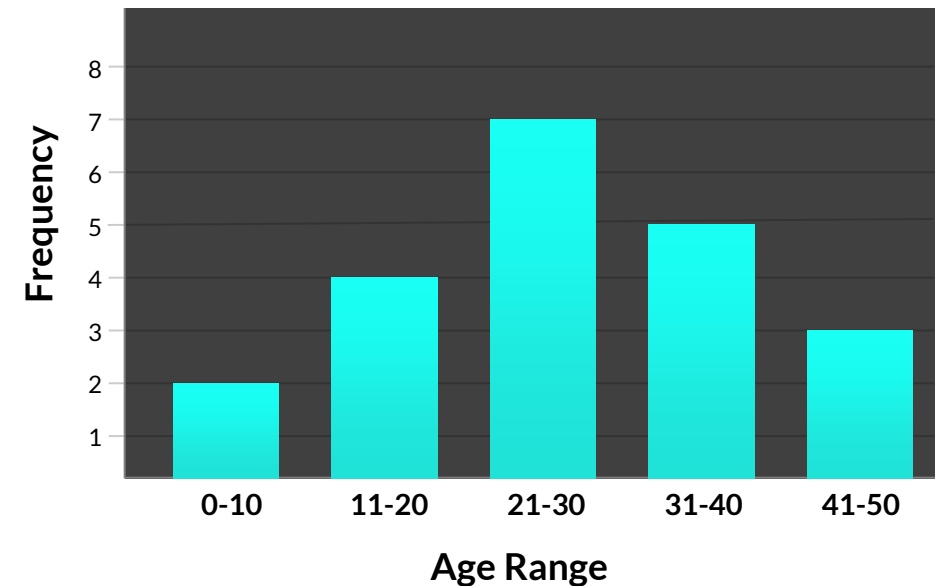
Data Profiling

Histograms are used to plot a single, discretized numerical variable

Imagine taking a numerical variable (like *age*), defining ranges or “bins” (1-5, 6-10, etc.), and counting the number of observations which fall into each bin; ***this is exactly what histograms are designed to do!***

Age Values:

8	29	45
25	33	37
19	43	21
28	32	40
24	17	28
5	22	39
15	47	12



KERNEL DENSITIES

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

Normal Distribution

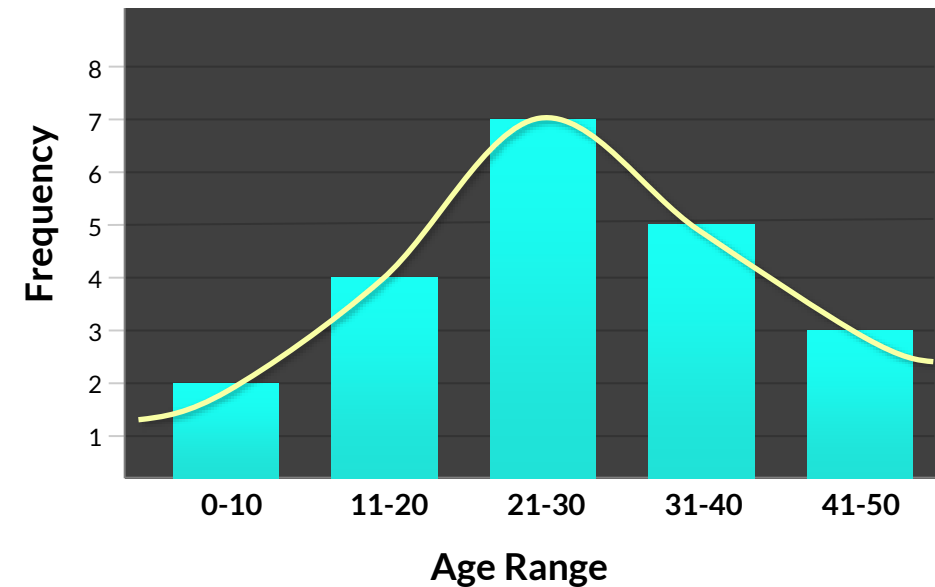
Data Profiling

Kernel densities are “smooth” versions of histograms, which can help to prevent users from over-interpreting breaks between bins

- **Technical definition:** “Non-parametric density estimation via smoothing”
- **Intuitive definition:** “Wet noodle laying on a histogram”

Age Values:

8	29	45
25	33	37
19	43	21
28	32	40
24	17	28
5	22	39
15	47	12



HISTOGRAMS & KERNEL DENSITIES

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

Normal Distribution

Data Profiling

When to use histograms and kernel density charts:

- Visualizing how a given variable is distributed
- Providing a visual glimpse of profiling metrics like **mean**, **mode**, and **skewness**

Things to watch out for:

- **Bin sensitivity:** Bin size can significantly change the shape and "smoothness" of a histogram, so select a bin width that accurately shows the data distribution
- **Outliers:** Histograms can be used to identify outliers in your data set, but you may need to remove them to avoid skewing the distribution
- **Sample size:** Histograms are best suited for variables with many observations, to reflect the true population distribution



PRO TIP: If your data is relatively symmetrical (*not skewed*), you can use **Sturge's Rule** as a quick "rule of thumb" to determine an appropriate number of bins: **$K = 1 + 3.322 \log(N)$** (where **K** = number of bins, **N** = number of observations)

CASE STUDY: HISTOGRAMS



THE SITUATION

You've just been promoted as the new Pit Boss at **The Lucky Roll Casino**. Your mission? Use data to help expose cheats on the casino floor.



THE ASSIGNMENT

Profits at the craps tables have been unusually low, and you've been asked to investigate the possible use of loaded die (*weighted towards specific numbers*). Your plan is to **track the outcome of each roll**, then **compare your results against the expected probability distribution** to see how closely they match.



THE OBJECTIVES

1. Record the outcomes for a series of individual dice rolls
2. Plot the frequency of each result (1-12) using a histogram
3. Compare your plot against the expected frequency distribution

NORMAL DISTRIBUTION

Categorical Variables

Categorical
Distributions

Numerical Variables

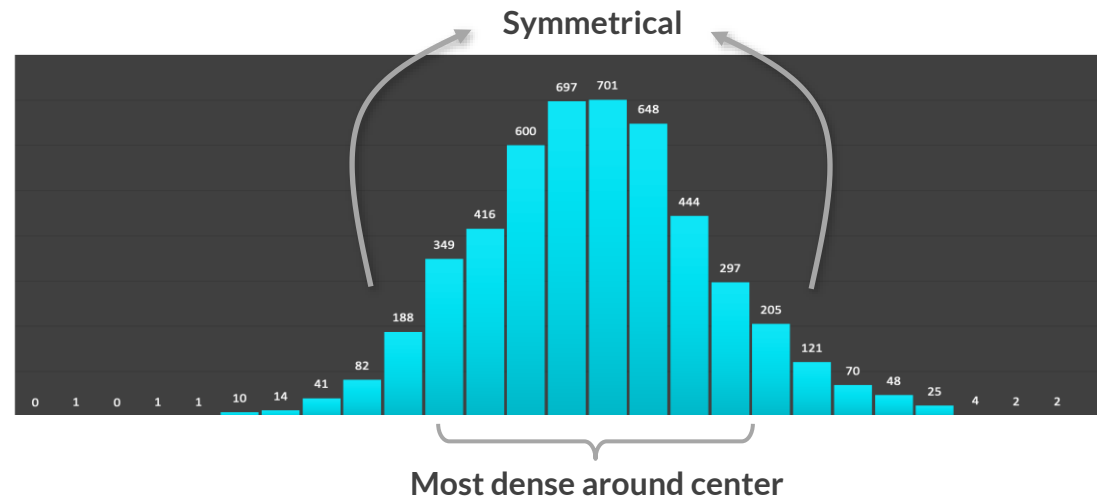
Histograms &
Kernel Densities

Normal Distribution

Data Profiling

Many numerical variables naturally follow a **normal distribution**, also known as a Gaussian distribution or “bell curve”

- Normal distributions are symmetrical and dense around the center, with flared out “tails” on both ends (*like a bell!*)
- Normal distributions are essential to many underlying assumptions in ML, and a helpful tool for comparing distributions or testing differences between them



You can find normal distributions in many real-world examples: **heights**, **weights**, **test scores**, etc.

NORMAL DISTRIBUTION

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

Normal Distribution

Data Profiling

The diagram illustrates the components of the Normal Distribution formula, $\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$. The formula is broken down into three colored boxes: a purple box for the coefficient $\frac{1}{\sigma\sqrt{2\pi}}$, a yellow box for the exponential base $e^{-\frac{1}{2}}$, and a cyan box for the squared term $\left(\frac{x-\mu}{\sigma}\right)^2$. Annotations with arrows explain each part: a purple arrow points from the coefficient box to the text 'Make the tails flare out'; a yellow arrow points from the exponential base box to the text 'Turn the parabola upside down'; and a cyan arrow points from the squared term box to the text 'Parabola centered around the mean'.

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Make the tails flare out

Turn the parabola upside down

Parabola centered around the mean

CASE STUDY: NORMAL DISTRIBUTION



THE SITUATION

It's August 2016, and you've been invited to Rio de Janeiro as a Data Analyst for the **Global Olympic Committee**.



THE ASSIGNMENT

Your job is to collect demographic data for all female athletes competing in the Summer Games and determine how the **distribution of Olympic athlete heights compares against the general public**.



THE OBJECTIVES

1. Gather heights for all female athletes competing in the 2016 Games
2. Plot height frequencies using a Histogram, and test various bin widths
3. Determine if athlete heights follow a normal distribution, or “bell curve”
4. Compare the distributions for athletes vs. the general public

DATA PROFILING

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

Normal Distribution

Data Profiling

Data profiling describes the process of using statistics to describe or summarize information about a particular variable

- Data profiling is critical for understanding variable characteristics which cannot be *seen* or easily visualized using tools like histograms
- Data profiling communicates **meaning**, using simple, concise, and universally understood metrics

Common data profiling metrics include:

- **Mode**
- **Mean**
- **Median**
- **Percentile**
- **Variance**
- **Standard Deviation**
- **Skewness**



PRO TIP: Combining visual distributions with data profiling metrics is a powerful way to communicate meaning in advanced analytics

MODE

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

Normal Distribution

Data Profiling

The **mode** is the most frequently observed value

- With a *numerical* variable (*most common*), it's the general range around the highest "peak" in the distribution
- With a *categorical* variable, it's simply the value that appears most often

Customer ID	First Name	Last Name	Gender	City	Sessions
1	Jennifer	Mcgrath	F	Seattle	24
2	Chad	Lewis	M	Seattle	12
3	Susan	Rodriguez	F	Houston	9
4	Wayne	Nielson	M	Miami	31
5	John	Depaul	M	Houston	24
6	Joseph	Martinez	M	Houston	11
7	Diane	Henry	F	Miami	3
8	Veronica	Comerford	F	Miami	24
9	Beverly	Nixon	F	Houston	15
10	Ivan	Layton	M	Houston	27

Mode of **City** = "Houston"

Mode of **Sessions** = 24

Mode of **Gender** = F, M
(this is a *bimodal* field!)

Common uses:

- Understanding the most common values within a dataset
- Diagnosing if one variable is influenced by another

MODE

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

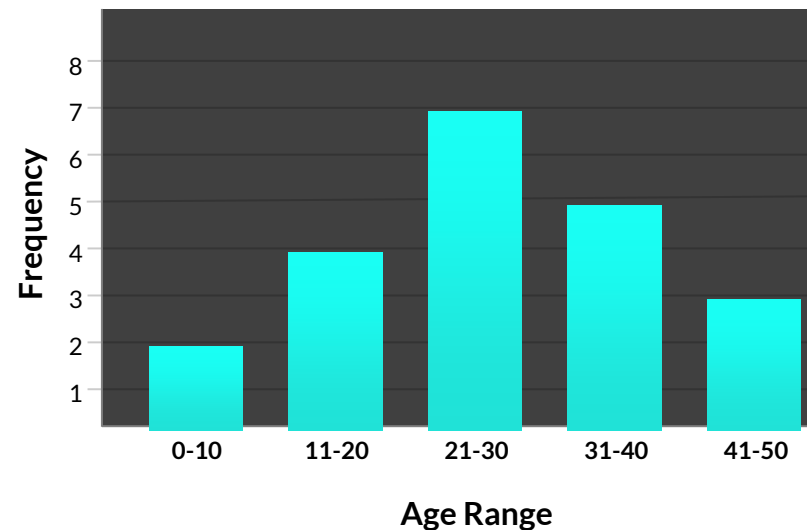
Normal Distribution

Data Profiling

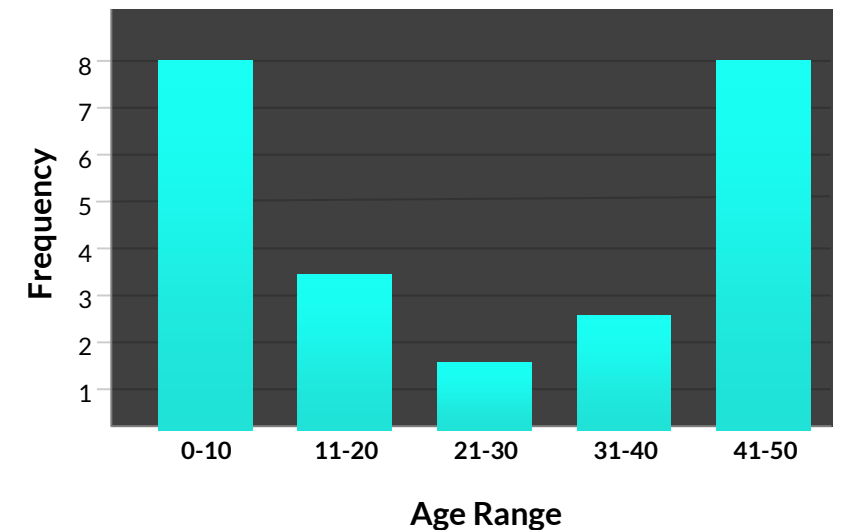
While **modes** typically aren't very useful on their own, they can provide helpful hints for deeper data exploration

- For example, the right histogram below shows a *multi-modal* distribution, which indicates that there may be another variable impacting the age distribution

Single Mode (**21-30**)



Two Modes (**0-10, 41-50**)



MEAN

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

Normal Distribution

Data Profiling

The **mean** is the calculated “central” value in a discrete set on numbers

- Mean is what most people think of when they hear the word “average”, and is calculated by dividing the **sum of all values** by the **count of all observations**
- Means can only be applied to *numerical* variables (*not categorical*)

Product Details				
ID	Size	Section	Price	Quantity
1	L	Camping	5	2000
2	L	Camping	10	1600
3	L	Camping	30	1100
4	L	Camping	100	500
5	L	Camping	200	20

$$\left. \begin{array}{c} \text{mean} = \frac{\text{sum of all values}}{\text{count of observations}} \\ = \frac{5,220}{5} = 1,044 \end{array} \right\}$$

Common uses:

- Making a “best-guess” estimate of a value
- Calculating a central value when outliers are not present

MEDIAN

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

Normal Distribution

Data Profiling

The **median** is the *middle value* in a list of values sorted from highest to lowest (*or vice versa*)

- When there are two middle-ranked values, the median is the **average** of the two
- Medians can only be applied to *numerical* variables (*not categorical*)

Customer ID	First Name	Last Name	Sessions
4	Wayne	Nielson	31
10	Ivan	Layton	27
1	Jennifer	Mcgrath	24
5	John	Depaul	24
8	Veronica	Comerford	24
9	Beverly	Nixon	15
2	Chad	Lewis	12
6	Joseph	Martinez	11
3	Susan	Rodriguez	9
7	Diane	Henry	3

} **Median = 19.5**
(average of 15 and 24)

Common uses:

- Identifying the “center” of a distribution
- Calculating a central value when outliers may be present

PERCENTILE

Categorical Variables

Categorical
Distributions

Numerical Variables

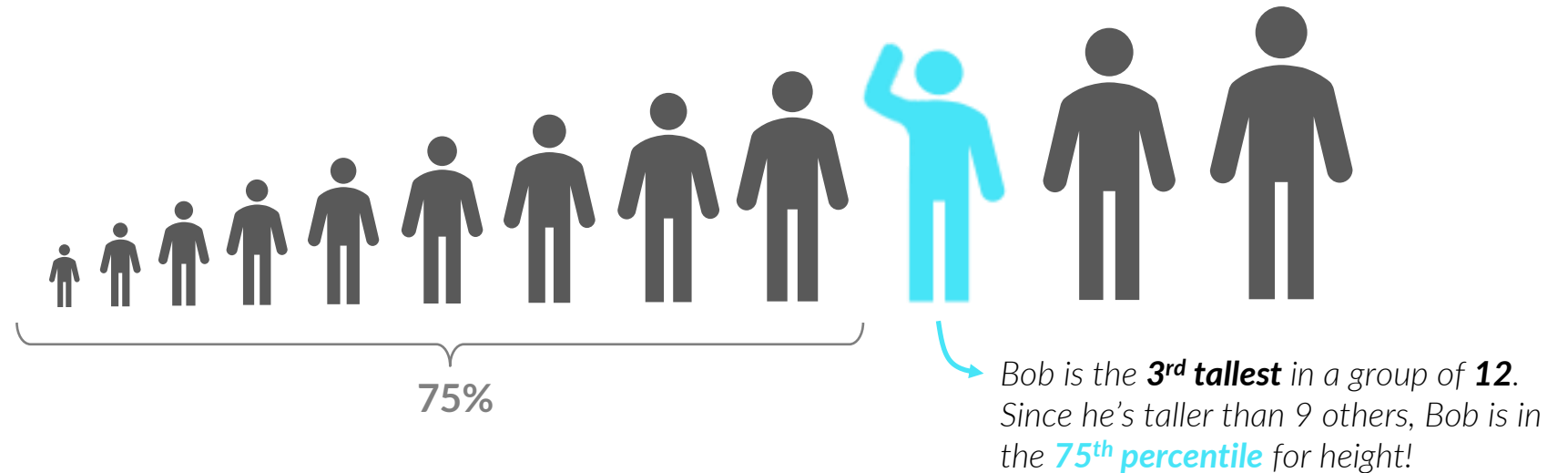
Histograms &
Kernel Densities

Normal Distribution

Data Profiling

Percentiles are used to describe the *percent of values* in a column which fall below a particular number

- If you are in the **90th** percentile for height, you're taller than **90%** of the population
- If the **50th** percentile for test scores is **86**, half of the students scored lower (*which means 86 is also the **median**!*)



Common uses:

- Providing context and intuitive benchmarks for how values “rank” within a sample (*test scores, height/weight, blood pressure, etc.*)

VARIANCE

Categorical Variables

Categorical
Distributions

Numerical Variables

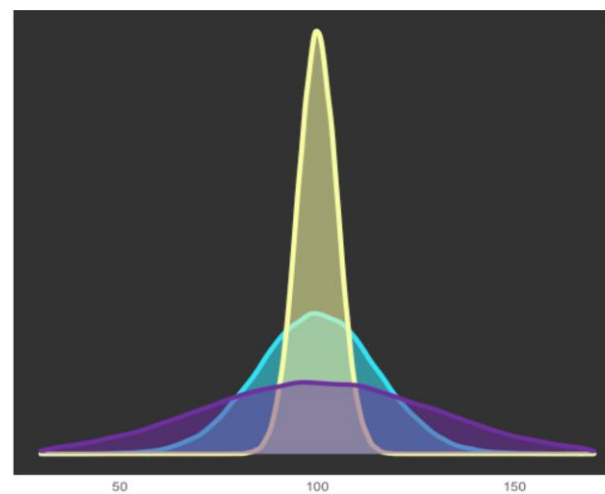
Histograms &
Kernel Densities

Normal Distribution

Data Profiling

Variance, in simple terms, describes how *thin* or *wide* a distribution is

- Variance measures how far the observations are from the mean, on average, and help us concisely describe a variable's distribution
- The *wider* a distribution, the *higher* the variance (*and vice versa*)



Variance = 5

Variance = 15

Variance = 30

Common uses:

- Comparing the numerical distributions of two different groups (*i.e. prices of products ordered online vs. in store*)

VARIANCE

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

Normal Distribution

Data Profiling

$$\frac{\sum_{i=1}^n (x_i - \mu)^2}{n - 1}$$

Calculation Steps:

- 1) Calculate the average of the variable
- 2) Subtract that average from the first row
- 3) Square that difference
- 4) Do steps 1-3 for every row and sum it up
- 5) Divide by the number of observations (-1)

***Average squared
distance from the mean***

STANDARD DEVIATION

Categorical Variables

Categorical
Distributions

Numerical Variables

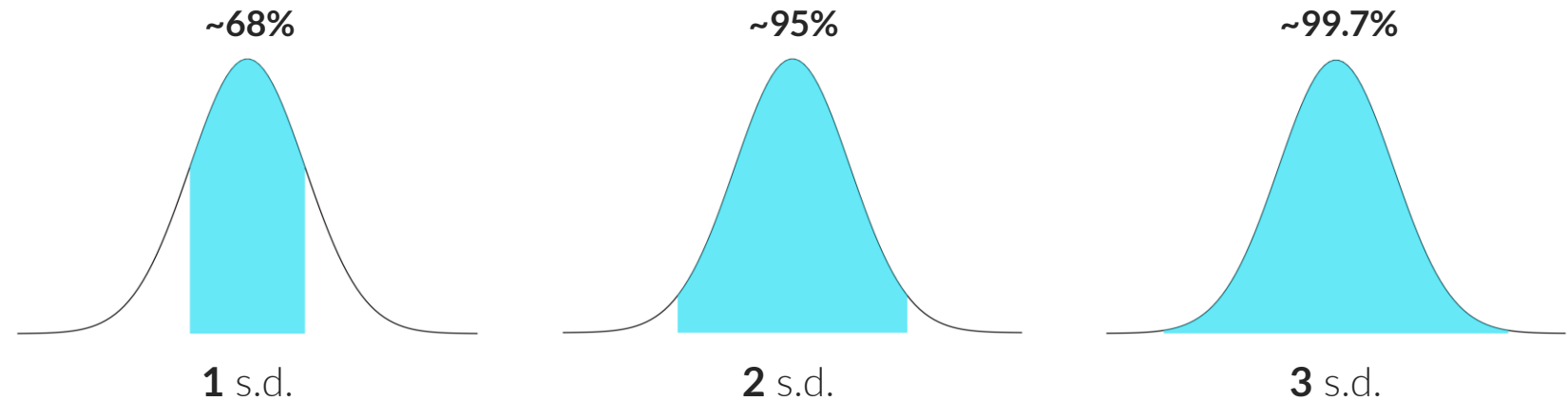
Histograms &
Kernel Densities

Normal Distribution

Data Profiling

Standard Deviation is the square root of the variance

- This converts the variance back to the scale of the variable itself
- In a normal distribution, **~68%** of values fall within 1 standard deviation of the mean, **~95%** fall within 2, and **~99.7%** fall within 3 (known as the “**empirical rule**”)



Common uses:

- Comparing segments for a given metric (*i.e. time on site for mobile users vs. desktop*)
- Understanding how likely certain values are bound to occur

SKEWNESS

Categorical Variables

Categorical
Distributions

Numerical Variables

Histograms &
Kernel Densities

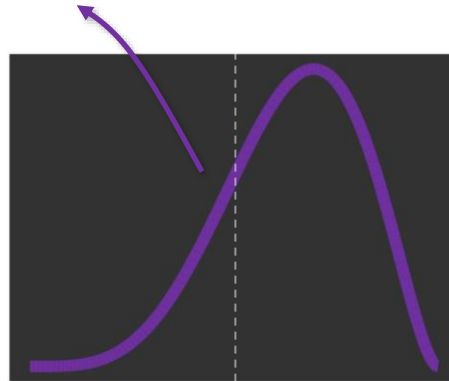
Normal Distribution

Data Profiling

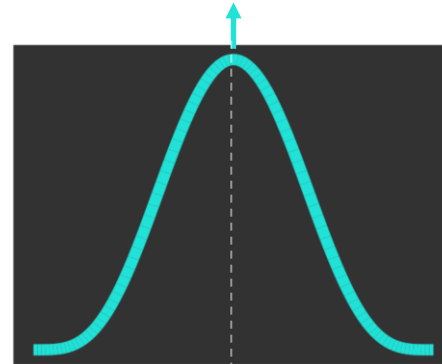
Skewness tells us how a distribution varies from a normal distribution

- This is commonly used to mathematically describe skew to the left or right

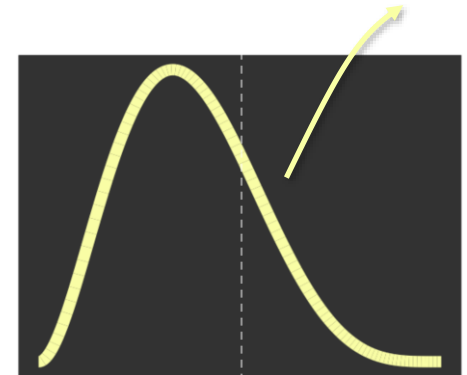
Left skew



Normal Distribution



Right skew



Common uses:

- Identifying non-normal distributions, and describing them mathematically

BEST PRACTICES: UNIVARIATE PROFILING



Make sure you are using the appropriate tools for profiling **categorical** variables vs. **numerical** variables



Distributions are a great way to quickly and visually explore variables



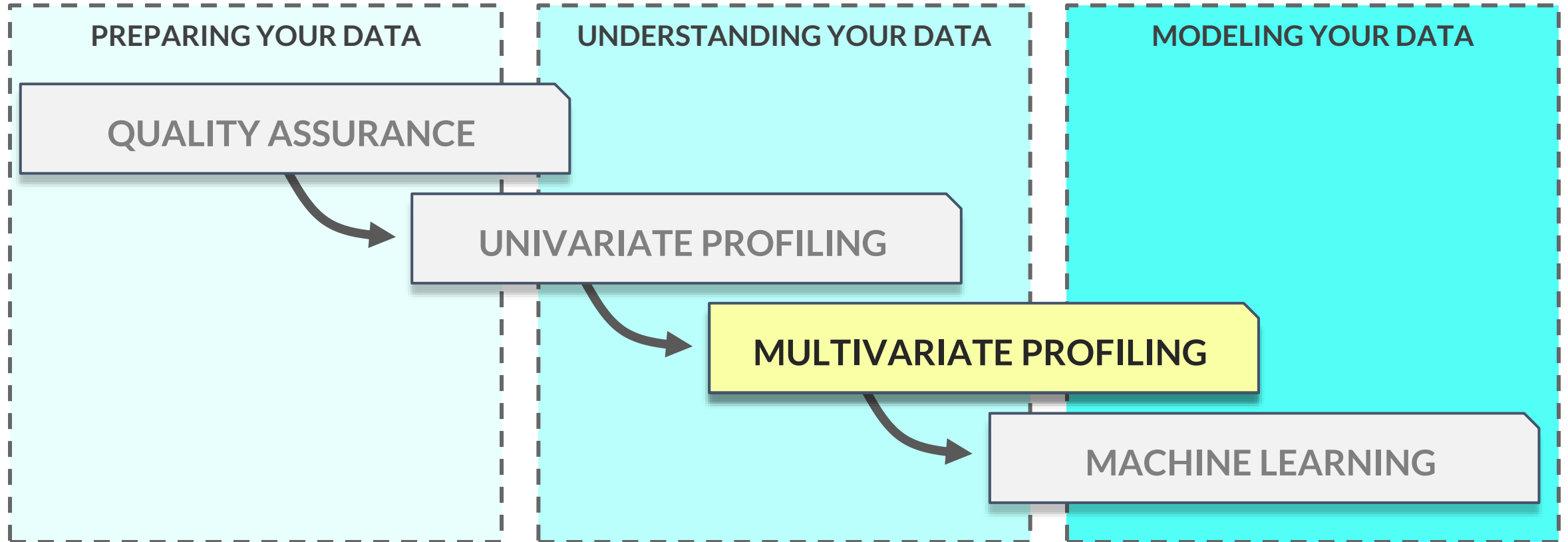
QA still comes first! Profiling metrics are important, but can lead to misleading results without proper QA (*i.e. handling outliers or missing values*)



One single distribution, visualization, or metric isn't enough to fully understand a variable; **always explore your data from multiple angles**

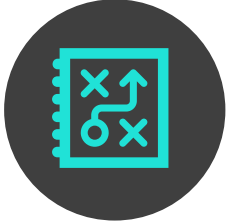
MULTIVARIATE DISTRIBUTIONS

MACHINE LEARNING PROCESS



Multivariate profiling is about **understanding relationships between *multiple* variables**. We'll cover common tools for exploring categorical & numerical data, including kernel densities, violin & box plots, scatterplots, etc.

MULTIVARIATE PROFILING



Multivariate profiling is the next step after univariate profiling, since single-metric distributions are rarely enough to draw meaningful insights or conclusions

TOPICS WE'LL COVER:

Categorical-Categorical
Distributions

Categorical-Numerical
Distributions

Multivariate Kernel
Densities

Violin & Box Plots

Numerical-Numerical
Distributions

Scatter Plots &
Correlation

COMMON USE CASES:

- Exploring relationships between multiple categorical and/or numerical variables
- Informing which specific machine learning models or techniques to use



Multivariate distributions are generally called “**joint**” **distributions**, and it’s clear why—it’s joining two variables into the same distribution!

CATEGORICAL-CATEGORICAL DISTRIBUTIONS

Categorical-Categorical
Distributions

Categorical-Numerical
Distributions

Multivariate Kernel
Densities

Violin & Box Plots

Numerical-Numerical
Distributions

Scatter Plots &
Correlation

Categorical/categorical distributions represent the frequency of *unique combinations* between two or more categorical variables

This is one of the simplest forms of multivariate profiling, and leverages the same tools we used to analyze univariate distributions:

- **Frequency tables:** Show the count (*or frequency*) of each distinct combination
- **Proportions tables:** Show the count of each combination as a % of the total
- **Heat maps:** Frequency or proportions table formatted to visualize patterns

Common uses:

- Understanding product mix based on multiple characteristics (*i.e. size & category*)
- Exploring customer demographics based on attributes like location, gender, etc.

CATEGORICAL-CATEGORICAL DISTRIBUTIONS

Categorical-Categorical
Distributions

Categorical-Numerical
Distributions

Multivariate Kernel
Densities

Violin & Box Plots

Numerical-Numerical
Distributions

Scatter Plots &
Correlation

In this example we're looking at product inventory using a distribution of two categorical fields: **Design** and **Size**

We can show this joint distribution as a simple count (*Frequency Table*), as a percentage of the total (*Proportions Table*), or as a conditionally formatted table (*Heat Map*)

Product Inventory					
	Design 1	Design 2	Design 3	Design 4	Design 5
X-Small	2	31	44	48	3
Small	5	33	39	33	19
Medium	5	48	28	36	37
Large	16	27	6	38	42
X-Large	35	28	1	31	3

} *Frequency Table*

Product Inventory					
	Design 1	Design 2	Design 3	Design 4	Design 5
X-Small	0.3%	4.9%	6.9%	7.5%	0.5%
Small	0.8%	5.2%	6.1%	5.2%	3.0%
Medium	0.8%	7.5%	4.4%	5.6%	5.8%
Large	2.5%	4.2%	0.9%	6.0%	6.6%
X-Large	5.5%	4.4%	0.2%	4.9%	0.5%

} *Proportions Table*

Product Inventory					
	Design 1	Design 2	Design 3	Design 4	Design 5
X-Small	2	31	44	48	3
Small	5	33	39	33	19
Medium	5	48	28	36	37
Large	16	27	6	38	42
X-Large	35	28	1	31	3

} *Heat Map*

CASE STUDY: HEAT MAPS



THE SITUATION

You've just been hired by the **New York Department of Transportation** (DOT) to help analyze traffic accidents in New York City from 2019-2020



THE ASSIGNMENT

The DOT commissioner would like to understand accident frequency by **time of day** and **day of week**, in order to support a public service campaign promoting safe driving habits.

Your role is to provide the data that she needs to understand **when traffic accidents are most likely to occur**.



THE OBJECTIVES

1. Create a table to plot accident frequency by time of day and day of week
2. Apply conditional formatting to the table to create a heatmap showing the days and times with the fewest (*green*) and most (*red*) accidents in the sample

CATEGORICAL-NUMERICAL DISTRIBUTIONS

Categorical-Categorical
Distributions

Categorical-Numerical
Distributions

Multivariate Kernel
Densities

Violin & Box Plots

Numerical-Numerical
Distributions

Scatter Plots &
Correlation

Categorical-numerical distributions are used for comparing numerical distributions across classes in a category (*i.e. age distribution by gender*)

These are typically visualized using variations of familiar univariate numerical distributions, including:

- **Histograms & Kernel Densities:** Show the count (*or frequency*) of values
- **Violin Plots:** Kernel density “glued” to its mirror image, and tilted on its side
- **Box Plots:** Like a kernel density, but formatted to visualize key statistical values (*min/max, median, quartiles*) and outliers

Common uses:

- Comparing key business metrics (*i.e. customer lifetime value, average order size, purchase frequency, etc.*) by customer class (*gender, loyalty status, location, etc.*)
- Comparing sales performance by day of week or hour of day

KERNEL DENSITIES

Categorical-Categorical
Distributions

Categorical-Numerical
Distributions

Multivariate Kernel
Densities

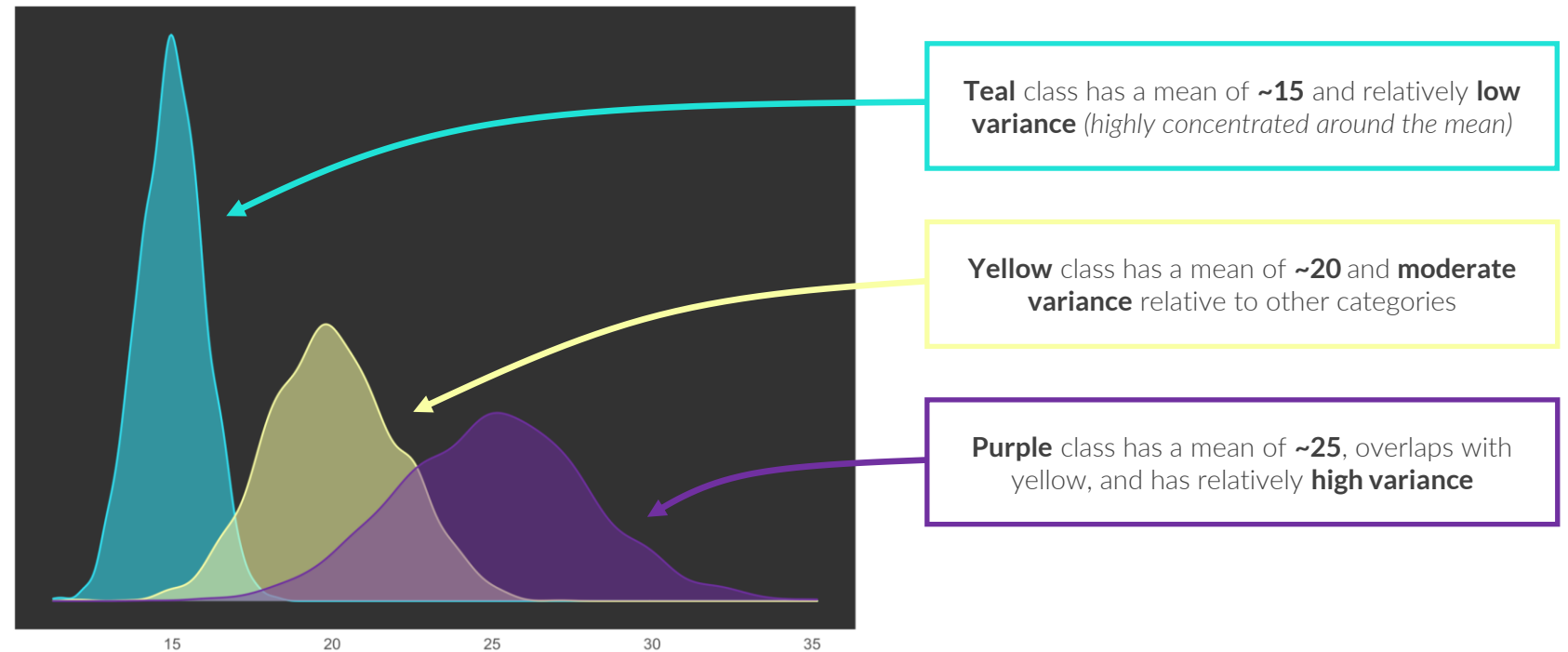
Violin & Box Plots

Numerical-Numerical
Distributions

Scatter Plots &
Correlation

Remember: **kernel densities** are just smooth versions of histograms

- To visualize a **categorical-numerical** distribution, kernel densities can be repeated to represent each class within a particular category



VIOLIN PLOTS

Categorical-Categorical
Distributions

Categorical-Numerical
Distributions

Multivariate Kernel
Densities

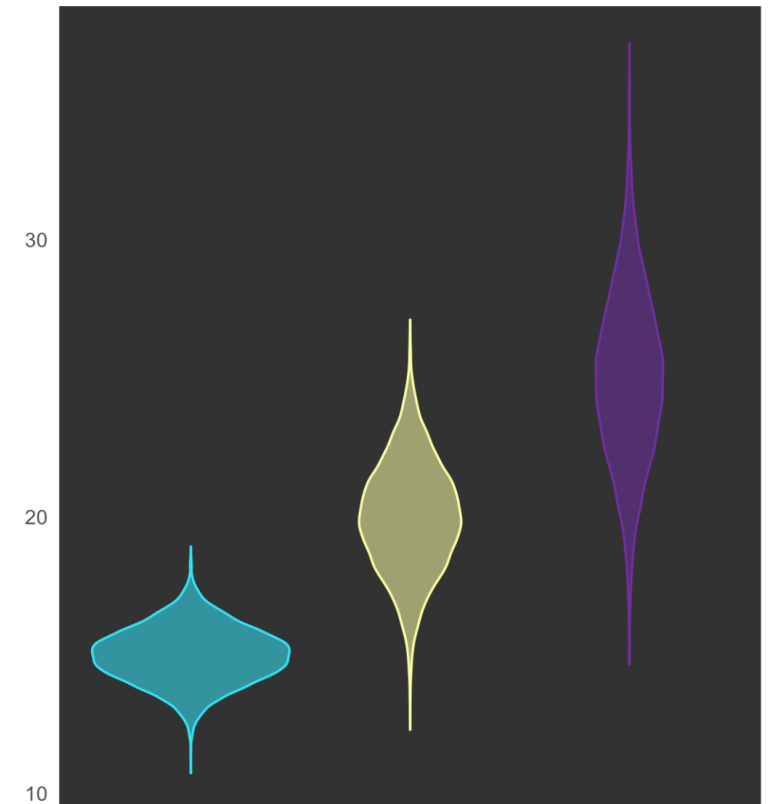
Violin & Box Plots

Numerical-Numerical
Distributions

Scatter Plots &
Correlation

A **violin plot** is essentially a kernel density flipped vertically and combined with its mirror image

- Violin plots use the exact same data as kernel densities, just visualized slightly differently
- These can help you visualize the shape of each distribution, and compare them across classes more clearly



BOX PLOTS

Categorical-Categorical
Distributions

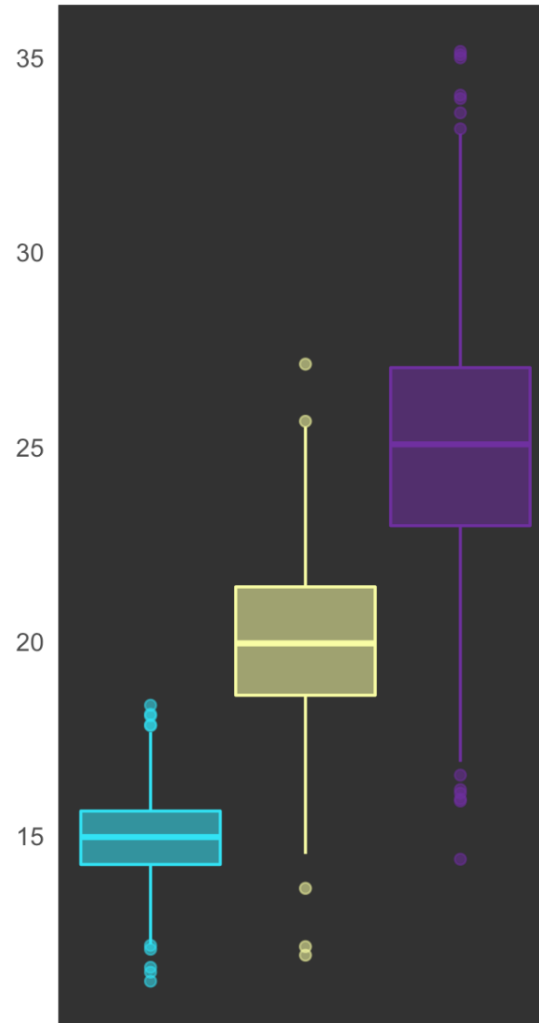
Categorical-Numerical
Distributions

Multivariate Kernel
Densities

Violin & Box Plots

Numerical-Numerical
Distributions

Scatter Plots &
Correlation



Box plots are like violin plots, but designed to show **key statistical attributes** rather than smooth distributions, including:

- **Median**
- **Min & Max** (*excluding outliers*)
- **25th & 75th Percentiles**
- **Outliers**

Box plots provide a *ton* of information in a single visual, and can be used to quickly compare statistical characteristics between classes

LIMITATIONS OF CATEGORICAL DISTRIBUTIONS

Categorical-Categorical
Distributions

Categorical-Numerical
Distributions

Multivariate Kernel
Densities

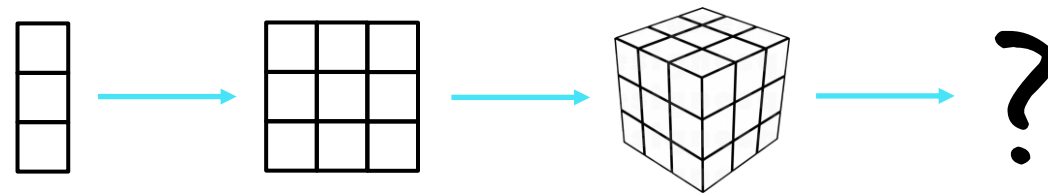
Violin & Box Plots

Numerical-Numerical
Distributions

Scatter Plots &
Correlation

Categorical profiling works for simple cases, but breaks down quickly

- Humans are pretty good at visualizing 1, 2, or maybe even 3 variables, but how would you visualize a joint distribution for **10** variables? **100**?



Categorical profiling can't answer **prescriptive** or **predictive** questions

- Suppose you randomized several elements on your sales page (*font, image, layout, button, copy, etc.*) to understand which ones drive conversions
- You could count conversions for individual elements, or some combinations of elements, but categorical distribution alone can't measure *causation*



This is when you need machine learning!

NUMERICAL-NUMERICAL DISTRIBUTIONS

Categorical-Categorical
Distributions

Categorical-Numerical
Distributions

Multivariate Kernel
Densities

Violin & Box Plots

Numerical-Numerical
Distributions

Scatter Plots &
Correlation

Numerical-Numerical distributions are common and intuitive, but the most complex mathematically

They are typically visualized using **scatter plots**, which plot points along the X and Y axis to show the relationship between two variables

- Scatter plots allow for simple, visual intuition: *when one variable increases or decreases, how does the other variable change?*
- There are many possibilities: no relationship, positive, negative, linear, non-linear, cubic, exponential, etc.

Common uses:

- Quickly visualizing how two numerical variables relate
- Predicting how a change in one variable will impact another (*i.e. square footage and house price, marketing spend and sales, etc.*)

CORRELATION

Categorical-Categorical
Distributions

Categorical-Numerical
Distributions

Multivariate Kernel
Densities

Violin & Box Plots

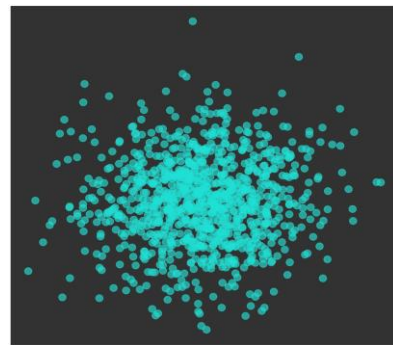
Numerical-Numerical
Distributions

Scatter Plots &
Correlation

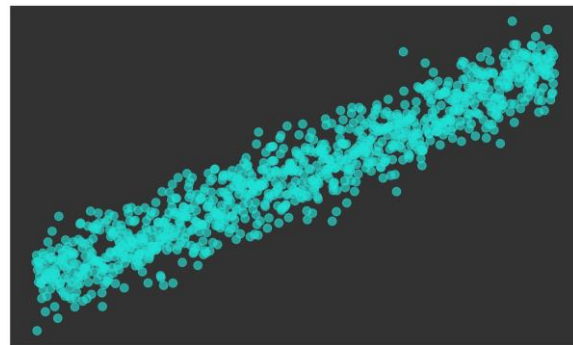
Univariate profiling metrics aren't much help in the multivariate world; we now need a way to describe *relationships* between variables

Correlation is the most common multivariate profiling metric, and is used to describe how a pair of variables are **linearly related**

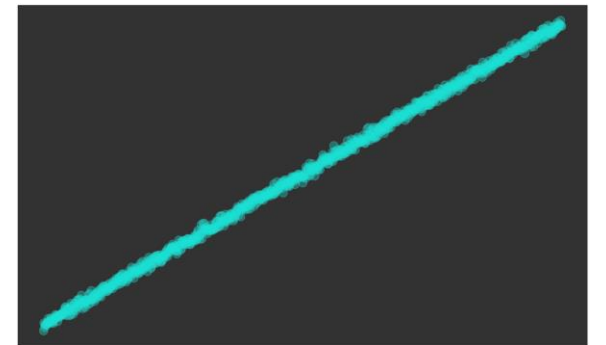
- **In other words:** *for a given row, when one variable's observation goes above its mean, does the other variable's observation also go above its mean (and vice versa)?*



No correlation



Positive correlation



Strong positive correlation

CORRELATION

Categorical-Categorical
Distributions

Categorical-Numerical
Distributions

Multivariate Kernel
Densities

Violin & Box Plots

Numerical-Numerical
Distributions

Scatter Plots &
Correlation

Correlation is an **extension of variance**

- Think of correlation as a way to measure the **variance** of both variables at one time (called “**co-variance**”), while controlling for the scales of each variable

Variance formula

$$\frac{\sum_{i=1}^n (x_i - \mu)^2}{n - 1}$$

Correlation formula

$$\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{(n - 1) s_x s_y}$$

- Here we multiply **variable X's difference from its mean** with **variable Y's difference from its mean**, instead of squaring a single variable (like we do with variance)
- s_x and s_y are the **standard deviations** of X and Y, which puts them on the same scale

CORRELATION VS. CAUSATION

Categorical-Categorical
Distributions

Categorical-Numerical
Distributions

Multivariate Kernel
Densities

Violin & Box Plots

Numerical-Numerical
Distributions

Scatter Plots &
Correlation

CORRELATION

! DOES **NOT** IMPLY !

CAUSATION

CORRELATION VS. CAUSATION

Categorical-Categorical
Distributions

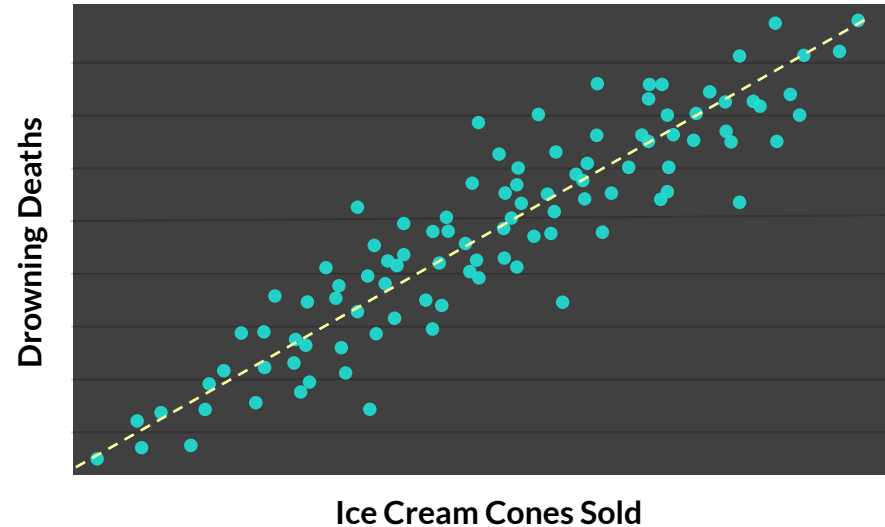
Categorical-Numerical
Distributions

Multivariate Kernel
Densities

Violin & Box Plots

Numerical-Numerical
Distributions

Scatter Plots &
Correlation



Consider the scatter plot above, showing daily **ice cream sales** and **drowning deaths** in a popular New England vacation town

- These two variables are clearly **correlated**, but do ice cream cones CAUSE people to drown? Do drowning deaths CAUSE a surge in ice cream sales?

Of course not, because correlation does NOT imply causation!

So what do you think is really going on here?

PRO TIP: VISUALIZING A THIRD DIMENSION

Categorical-Categorical
Distributions

Categorical-Numerical
Distributions

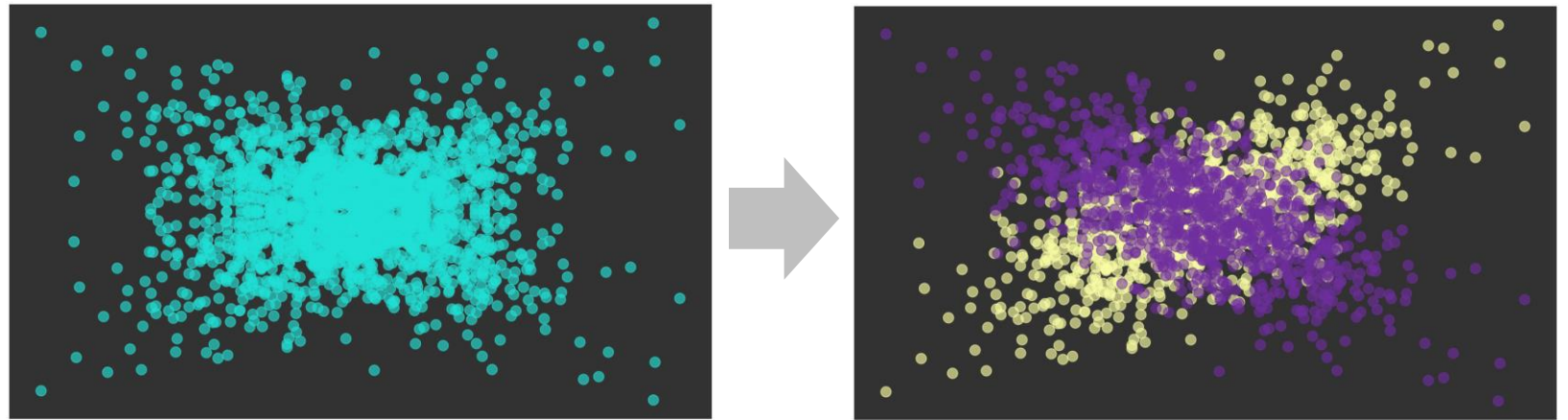
Multivariate Kernel
Densities

Violin & Box Plots

Numerical-Numerical
Distributions

Scatter Plots &
Correlation

Scatter plots show two dimensions by default (*X and Y*), but using **symbols** or **color** allows you to visualize additional variables and expose otherwise hidden patterns or trends



Visualizing more than 3-4 dimensions is beyond human capability.
This is where you need machine learning!

CASE STUDY: CORRELATION



THE SITUATION

You've just landed your dream job as a Marketing Analyst at **Loud & Clear**, the hottest ad agency in San Diego.



THE ASSIGNMENT

Your client would like to understand the **impact of their digital media spend**, and how it relates to website traffic, offline spend, site load time, and sales.

Your role is to **collect and visualize these metrics** at the weekly-level in order to begin exploring the relationships between them.



THE OBJECTIVES

1. Gather weekly spend, traffic, website, and sales data
2. Create a scatterplot to visualize any two given variables
3. Compare the relationship between Digital Media Spend and each variable
4. What patterns do you see, and how would you interpret them?

BEST PRACTICES: MULTIVARIATE PROFILING



Univariate profiling is a great start, but multivariate profiling is necessary when working with **more than one variable** (*pretty much always*)



Understand the **types of variables** you're working with (*categorical vs. numerical*) to determine which profiling and visualization techniques to use



Use categorical variables to **filter or “cut”** your data and quickly compare profiling metrics or distributions across classes



Remember that **correlation does not imply causation**, and that variables can be related without one *causing* a change in the other

LOOKING AHEAD

CONGRATULATIONS!

Now that you've completed **Part 1: QA & Data Profiling**, you should have a strong grasp of QA techniques, univariate & multivariate distributions, and common data profiling metrics.

In **Part 2** we'll dive into **supervised machine learning** and explore powerful classification techniques like K-Nearest Neighbors, Naïve Bayes, Decision Trees, Logistic Regression, Sentiment Analysis and more.



PART 1

QA & Data Profiling



PART 2

Classification



PART 3

Regression & Forecasting



PART 4

Unsupervised Learning